POWER BI BASICS

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Question 1: What is Power BI and why is it used in businesses?

Power BI is a business analytics service by Microsoft that provides interactive visualizations and business intelligence capabilities. It is used in businesses to:

- **Transform raw data** into meaningful insights and interactive reports.
- **Make data-driven decisions** by visualizing trends and patterns across various departments.
- **Consolidate data** from multiple sources into a single dashboard for real-time monitoring.

Question 2: Name and explain the three main components of Power BI.

The three primary components are:

- **Power BI Desktop:** A free Windows application used to connect to, transform, and visualize your data through report authoring.
- **Power BI Service:** A cloud-based (SaaS) service where users can publish, share, and collaborate on reports and dashboards.
- **Power BI Mobile:** Apps for iOS, Android, and Windows devices used to view and interact with reports on the go.

Question 3: Explain the Power BI workflow.

The standard workflow follows these steps:

1. **Bring data into Power BI Desktop** and create a report.
2. **Publish the report** to the Power BI Service.
3. **Share the report** with others so they can view and interact with the data in the Service or on Mobile apps.

Question 4: List any four data cleaning tasks that can be performed in Power Query.

Within the Power Query Editor, you can:

- **Remove Rows:** Eliminate duplicates or null/empty rows.
- **Change Data Types:** Convert columns to Text, Date, or Decimal numbers.
- **Split Columns:** Divide a single column into multiple columns based on a delimiter (e.g., a comma or space).
- **Replace Values:** Search for specific values and replace them with something else.

Question 5: Write step by step instructions to load the Real_Estate dataset.

To load the dataset:

1. Open **Power BI Desktop**.
2. Click on **"Get Data"** in the Home ribbon.
3. Select the file format (e.g., **Excel** or **Text/CSV**) that matches your Real_Estate file.
4. Browse and **select the file**, then click **Open**.
5. In the Navigator window, check the box next to the table/sheet name and click **"Load"** (or "Transform Data" if cleaning is required).

Question 6: Define Data View, Report View, and Model View.

- **Report View:** The canvas where you create visualizations, charts, and maps to build your report.
- **Data View:** A view that shows your data in a tabular format, allowing you to inspect rows and create calculated columns or measures.
- **Model View:** A view used to manage relationships between different tables in your data model.

Question 7: Discuss the different data sources that Power BI supports.

Power BI supports a wide variety of sources:

- **Files:** Excel, CSV, XML, JSON, and PDF.
- **Databases:** SQL Server, Oracle, MySQL, and PostgreSQL.
- **Cloud/Online Services:** Azure SQL, SharePoint, Google Analytics, and Salesforce.
- **Web:** Pulling data directly from a website URL.

Question 8: Split Owner Name to create two new columns as First Name and Last Name.

To perform this in Power Query:

1. Right-click the **"Owner Name"** column.
2. Select **"Split Column"** -> **"By Delimiter"**.
3. Choose **"Space"** as the delimiter and click OK.
4. Rename the resulting columns to **"First Name"** and **"Last Name"**.